

Suppose that $X_1, X_2, \dots, X_n$ is a random sample from the exponential distribution with rate $\lambda > 0$.

Derive a hypothesis test of size α for

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda > \lambda_0$$

What statistic should we use?

Test 1: Using the Sample Mean

Step One:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Choose a statistic.

$\bar{X}$

Test 1: Using the Sample Mean

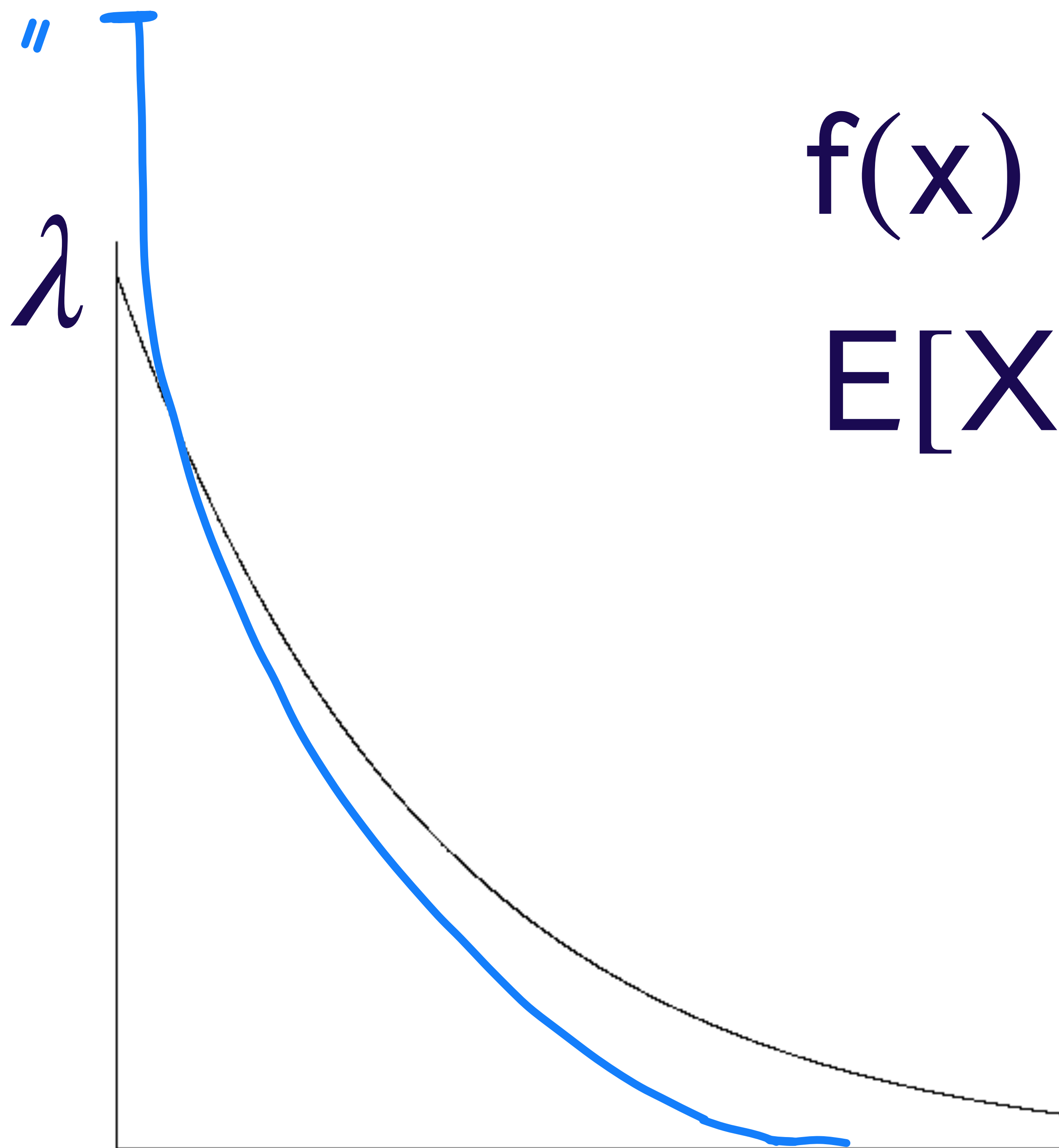
Step Two:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Give the form of the test.

"larger λ "



$$f(x) = \lambda e^{-\lambda x}$$

$$E[X] = 1/\lambda$$

Test 1: Using the Sample Mean

Step Two:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Give the form of the test.

Reject H_0 , in favor of H_1 , if

$$\bar{X} < c$$

for some c to be determined.

Test 1: Using the Sample Mean

Step Three:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Find c .

$$\alpha = P(\text{Type I Error})$$

$$= P(\text{Reject } H_0; \lambda_0)$$

$$= P(\bar{X} < c; \lambda_0)$$

Test 1: Using the Sample Mean

Step Three:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Find c.

$$\alpha = P(\bar{X} < c; \lambda_0)$$

$$= P(2n\lambda_0\bar{X} < 2n\lambda_0c; \lambda_0)$$

$$= P(W < 2n\lambda_0c; \lambda_0)$$

where $W \sim \chi^2(2n)$.

Test 1: Using the Sample Mean

Step Three:

$$H_0 : \lambda = \lambda_0$$

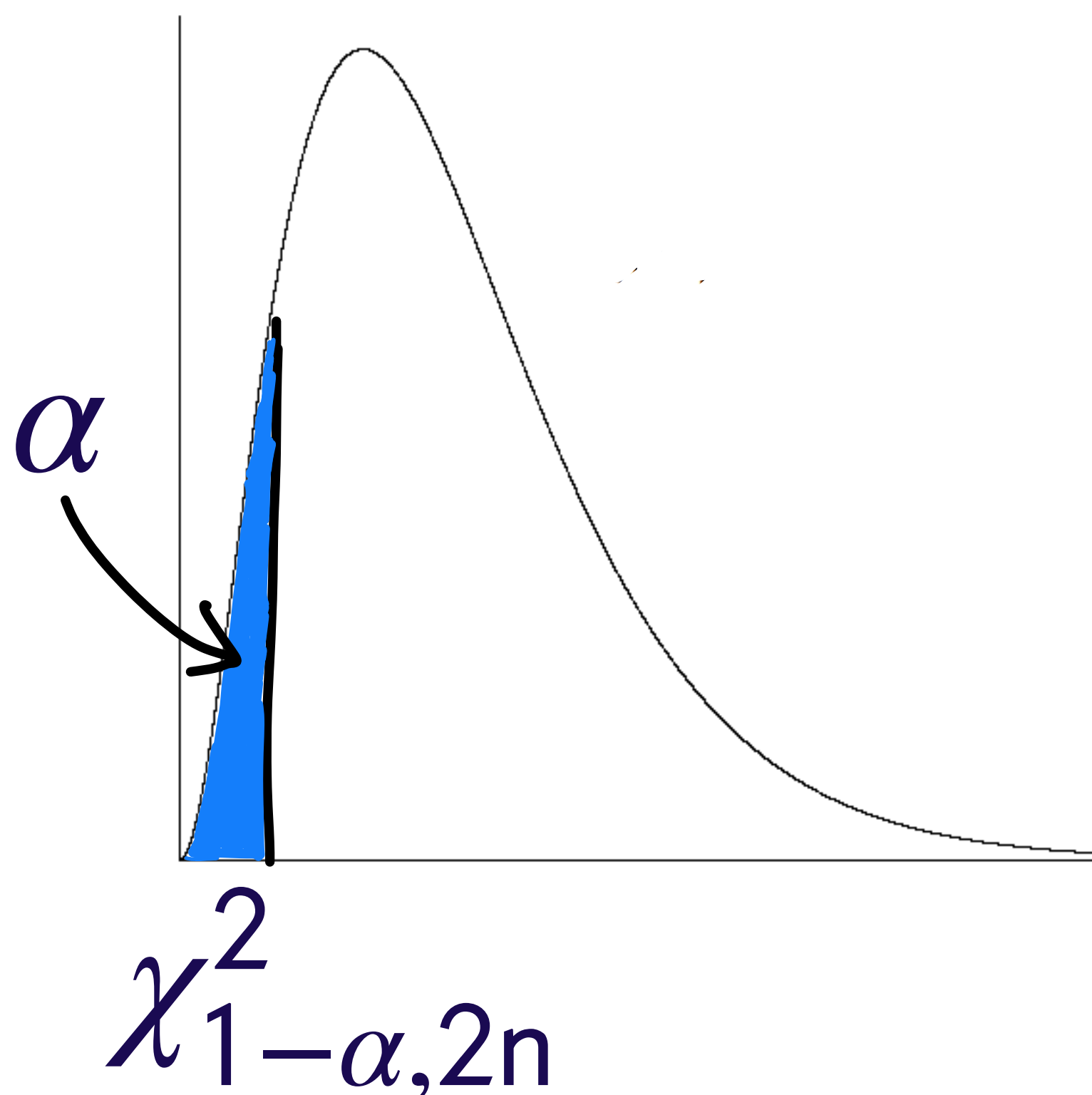
$$H_1 : \lambda > \lambda_0$$

Find c.

$$\alpha = P(W < 2n\lambda_0 c;)$$

We want

$$2n\lambda_0 c = \chi^2_{1-\alpha, 2n}$$



Test 1: Using the Sample Mean

Step Four:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Conclusion.

Reject H_0 , in favor of H_1 , if

$$\bar{X} < \frac{\chi^2_{1-\alpha, 2n}}{2n\lambda_0}$$

$$\chi^2_{\alpha,n}$$

In R, get

$$\chi^2_{0.10,6}$$

by typing

`qchisq(0.90,6)`

Test 2: Using the Sample Minimum

Step One:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Choose a statistic.

$$Y_n = \min(X_1, X_2, \dots, X_n)$$

Test 2: Using the Sample Minimum

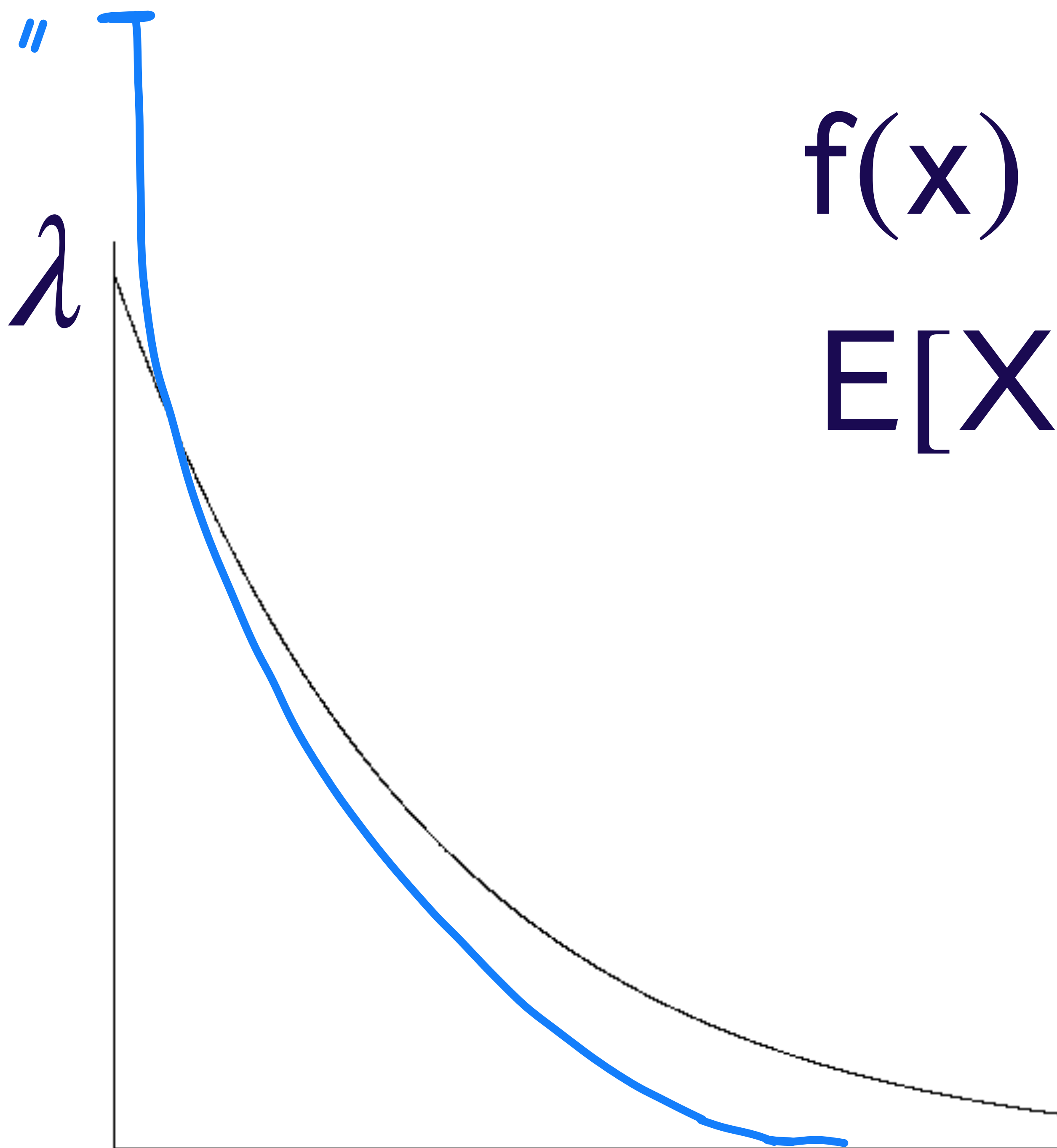
Step Two:

$$H_0 : \lambda = \lambda_0$$

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Give the form of the test.

"larger λ "



$$f(x) = \lambda e^{-\lambda x}$$

$$E[X] = 1/\lambda$$

Test 2: Using the Sample Minimum

Step Two:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Give the form of the test.

Reject H_0 , in favor of H_1 , if

$$Y_n < c$$

for some c to be determined.

Test 2: Using the Sample Minimum

Step Three:

$$H_0 : \lambda = \lambda_0$$

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Find c .

$$\alpha = P(\text{Type I Error})$$

$$= P(\text{Reject } H_0; \lambda_0)$$

$$= P(Y_n < c; \lambda_0)$$

Test 2: Using the Sample Minimum

Step Three:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Find c.

$$\alpha = P(Y_n < c; \lambda_0)$$

$$Y_n \sim \exp(\text{rate} = n\lambda_0)$$

$$n\lambda_0 Y_n \sim \exp(\text{rate} = 1)$$

Test 2: Using the Sample Minimum

Step Three:

$$H_0 : \lambda = \lambda_0$$

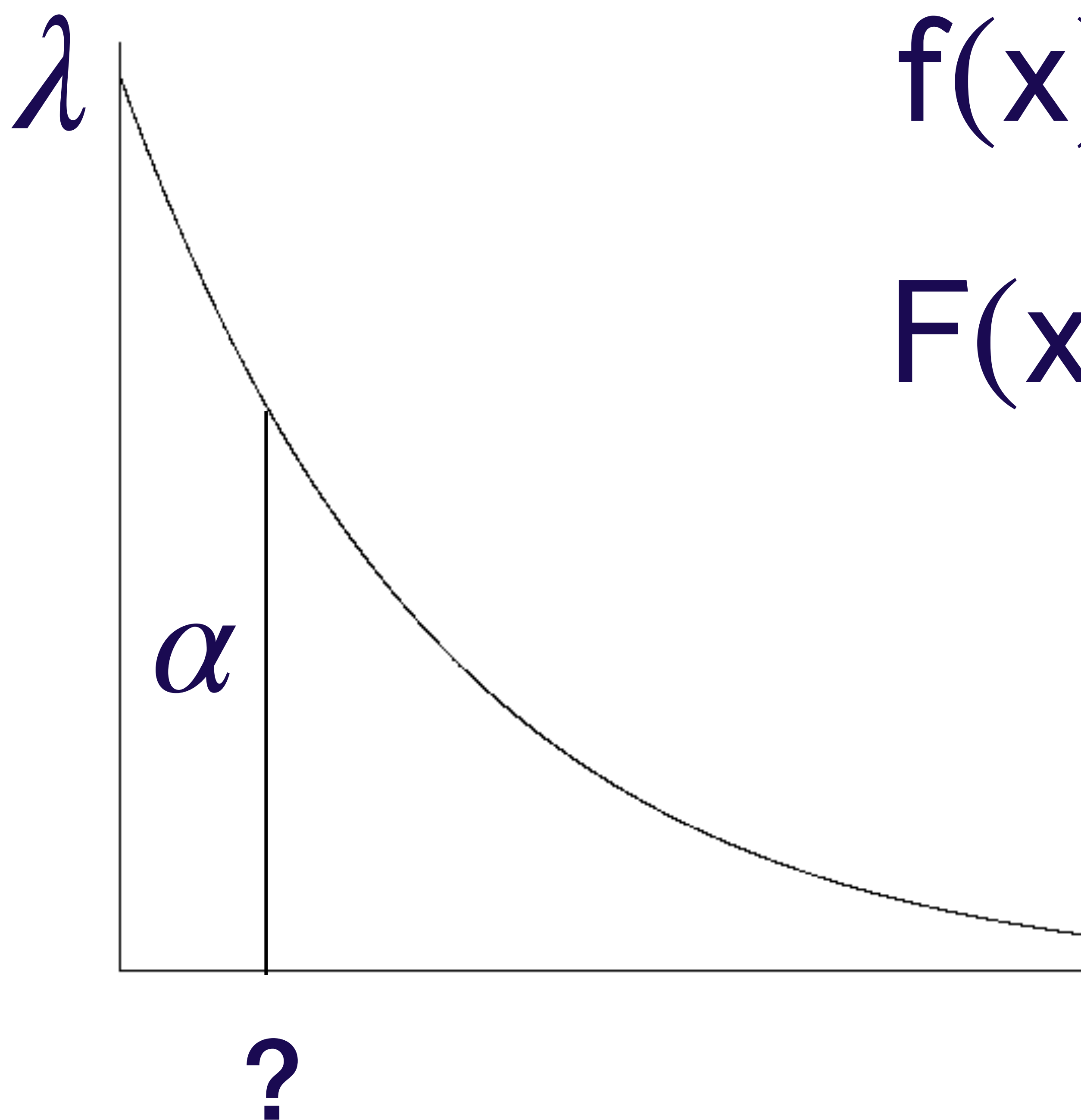
$$H_1 : \lambda > \lambda_0$$

Find c .

$$\alpha = P(n\lambda_0 Y_n < cn\lambda_0; \lambda_0)$$

$$= P(X < cn\lambda_0)$$

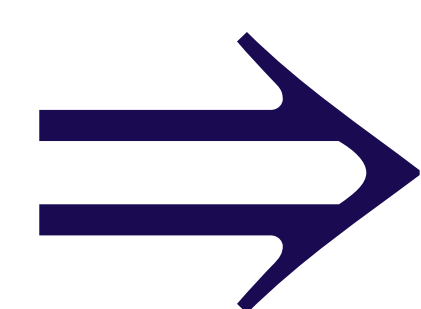
where $X \sim \exp(\text{rate} = 1)$.



$$f(x) = e^{-x}$$

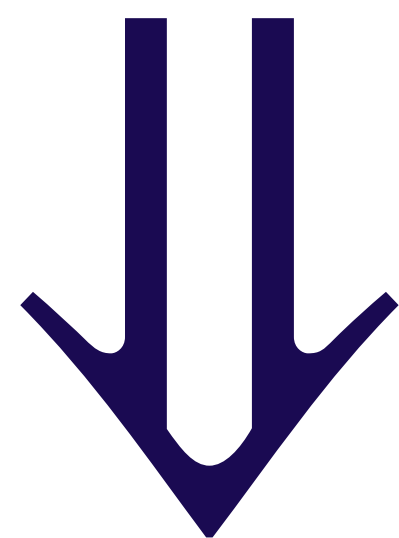
$$F(x) = P(X \leq x) \\ = 1 - e^{-x}$$

$$1 - e^{-?} = \alpha$$

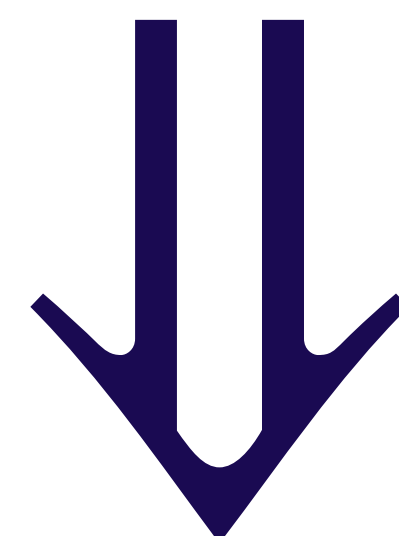


$$? = -\ln(1 - \alpha)$$

$$\alpha = P(X < cn\lambda_0)$$



$$cn\lambda_0 = -\ln(1 - \alpha)$$



$$c = \frac{-\ln(1 - \alpha)}{n\lambda_0}$$

Test 2: Using the Sample Minimum

Step Four:

$$H_0 : \lambda = \lambda_0$$

$$H_1 : \lambda > \lambda_0$$

Conclusion.

Reject H_0 , in favor of H_1 , if

$$Y_n = \min(X_1, X_2, \dots, X_n) < \frac{-\ln(1 - \alpha)}{n\lambda_0}$$

Compare the Tests:

Test 1: based on $\bar{X}$

$$\gamma_1(\lambda) = P(\text{Reject } H_0 ; \lambda)$$

$$= P\left(\bar{X} < \frac{\chi^2_{1-\alpha, 2n}}{2n\lambda_0} ; \lambda\right)$$

$$= P\left(2n\lambda\bar{X} < 2n\lambda\frac{\chi^2_{1-\alpha, 2n}}{2n\lambda_0} ; \lambda\right)$$

$$= P\left(W < \frac{\lambda}{\lambda_0}\chi^2_{1-\alpha, 2n}\right) \quad W \sim \chi^2(2n)$$

$$n = 10 \quad \lambda_0 = 1 \quad \alpha = 0.05$$

$$\chi^2_{1-\alpha, 2n} = \chi^2_{0.95, 20} = 10.851$$

$$\text{qchisq}(0.05, 20)$$

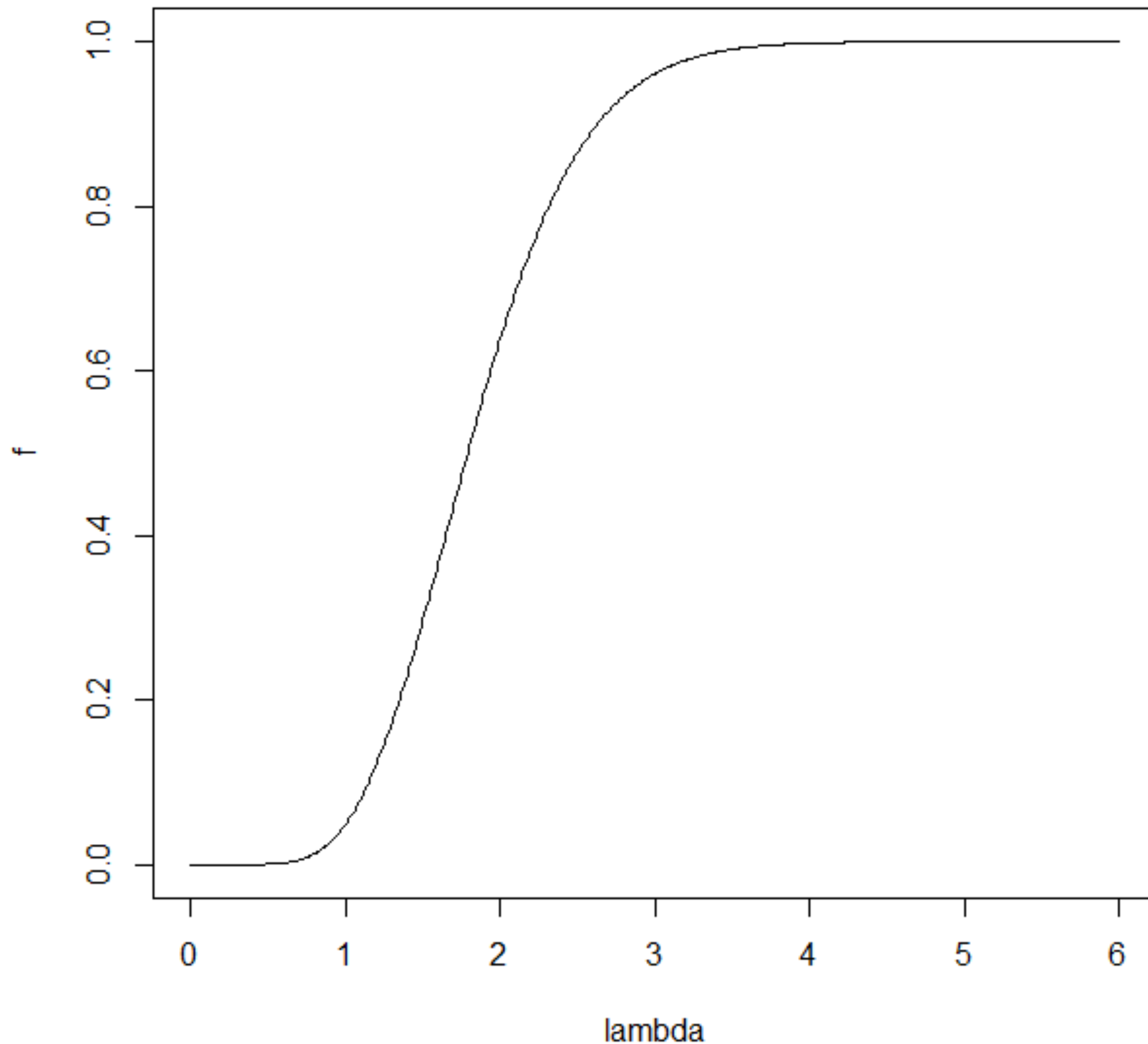
$$\gamma_1(\lambda) = P\left(W < \frac{\lambda}{\lambda_0} \chi^2_{1-\alpha, 2n}\right)$$

$$= P(W < 10.851\lambda)$$

Plot in R:

```
lambda<-seq(0,6,0.01)
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f<-pchisq(10.81*lambda,20)
```



Compare the Tests:

Test 2: based on

$$Y_n = \min(X_1, X_2, \dots, X_n)$$

$$\gamma_2(\lambda) = P(\text{Reject } H_0 ; \lambda)$$

$$= P\left(Y_n < \frac{-\ln(1-\alpha)}{n\lambda_0} ; \lambda\right)$$

$$= 1 - e^{-n\lambda(-\ln(1-\alpha)/n\lambda_0)}$$

$$= 1 - e^{-\lambda(-\ln(1-\alpha)/\lambda_0)}$$

exp(rate = nλ)

Compare the Tests:

Test 2: based on

$$Y_n = \min(X_1, X_2, \dots, X_n)$$

$$\gamma_2(\lambda) = P(\text{Reject } H_0 ; \lambda)$$

$$= P\left(Y_n < \frac{-\ln(1 - \alpha)}{n\lambda_0} ; \lambda\right)$$

$$= 1 - e^{-n\lambda(-\ln(1 - \alpha)/n\lambda_0)}$$

$$= 1 - e^{-\lambda(-\ln(1 - \alpha)/\lambda_0)}$$

Compare the Tests:

Test 2: based on

$$\gamma_2(\lambda) = 1 - e^{-\lambda(-\ln(1-\alpha)/\lambda_0)}$$

$$= 1 - (1 - \alpha)^{\lambda/\lambda_0}$$

$$H_0 : \lambda = 1$$

$$H_1 : \lambda > 1$$

